The type of small ordinals is an ordinal in the next universe

The Univalent Foundations Program

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Editorial note. Original author statement, proof and supporting context preserved. Only the document wrapper and navigational reference bindings are added; no mathematical repair or correctness certification.

Allowed standard prerequisites (editorial).

- Homotopy type theory. Univalence, dependent products and sums, identity types, propositional and n-truncation; native book references retained.. Standard background, without replacing the counted argument.

Source reference. Original reference 5.7: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1660–1660. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 10.1.2: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1670–1670. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 9.8: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1706–1706. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 5.3: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1724–1724. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 3.5: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1735–1735. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

10.1 Ordinal numbers

Definition 10.1.1. Let A be a set and

$$(- < -) : A \to A \to \mathsf{Prop}$$

a binary relation on A. We define by induction what it means for an element $a : A$ to be **accessible** by $<$:

- If b is accessible for every $b < a$, then a is accessible.

We write $\mathsf{acc}(a)$ to mean that a is accessible.

It may seem that such an inductive definition can never get off the ground, but of course if a has the property that there are *no* b such that $b < a$, then a is vacuously accessible.

Note that this is an inductive definition of a family of types, like the type of vectors considered in original reference 5.7. More precisely, it has one constructor, say $\mathsf{acc}_<$, with type

$$\mathsf{acc}_< : \prod_{a:A} \Big(\prod_{b:A} (b < a) \to \mathsf{acc}(b) \Big) \to \mathsf{acc}(a).$$

The induction principle for acc says that for any $P : \prod_{(a:A)} \mathsf{acc}(a) \to \mathcal{U}$, if we have

$$f : \prod_{(a:A)} \prod_{(h:\prod_{(b:A)}(b<a)\to\mathsf{acc}(b))} \Big(\prod_{(b:A)} \prod_{(l:b<a)} P(b, h(b,l)) \Big) \to P(a, \mathsf{acc}_<(a,h)),$$

then we have $g : \prod_{(a:A)} \prod_{(c:\mathsf{acc}(a))} P(a,c)$ defined by induction, with

$$g(a, \mathsf{acc}_<(a,h)) \equiv f(a,\ h,\ \lambda b.\, \lambda l.\, g(b, h(b,l))).$$

This is a mouthful, but generally we apply it only in the simpler case where $P : A \to \mathcal{U}$ depends only on A. In this case the second and third arguments of f may be combined, so that what we have to prove is

$$f : \prod_{a:A} \Big(\prod_{b:A} (b < a) \to \mathsf{acc}(b) \times P(b) \Big) \to P(a).$$

That is, we assume every $b < a$ is accessible and $g(b) : P(b)$ is defined, and from these define $g(a) : P(a)$.

The omission of the second argument of P is justified by the following lemma, whose proof is the only place where we use the more general form of the induction principle.

Lemma 10.1.2. *Accessibility is a mere property.*

Proof. We must show that for any $a : A$ and $s_1, s_2 : \mathsf{acc}(a)$ we have $s_1 = s_2$. We prove this by induction on s_1, with

$$P_1(a, s_1) :\equiv \prod_{s_2:\mathsf{acc}(a)} (s_1 = s_2).$$

Thus, we must show that for any $a : A$ and $h_1 : \prod_{(b:A)}(b < a) \to \mathsf{acc}(b)$ and

$$k_1 : \prod_{(b:A)} \prod_{(l:b<a)} \prod_{(t:\mathsf{acc}(b))} h_1(b,l) = t,$$

we have $\mathsf{acc}_<(a,h) = s_2$ for any $s_2 : \mathsf{acc}(a)$. We regard this statement as $\prod_{(a:A)} \prod_{(s_2:\mathsf{acc}(a))} P_2(a, s_2)$, where

$$P_2(a, s_2) :\equiv \prod_{(h_1:\cdots)} \prod_{(k_1:\cdots)} (\mathsf{acc}_<(a, h_1) = s_2);$$

thus we may prove it by induction on s_2. Therefore, we assume $h_2 : \prod_{(b:A)}(b < a) \to \mathsf{acc}(b)$, and k_2 with a monstrous but irrelevant type, and must show that for any h_1 and k_1 with types as above, we have $\mathsf{acc}_<(a, h_1) = \mathsf{acc}_<(a, h_2)$. By function extensionality, it suffices to show $h_1(b, l) = h_2(b, l)$ for all $b : A$ and $l : b < a$. This follows from k_1. □

Definition 10.1.3. A binary relation $<$ on a set A is **well-founded** if every element of A is accessible.

The point of well-foundedness is that for $P : A \to \mathcal{U}$, we can use the induction principle of acc to conclude $\prod_{(a:A)} \mathsf{acc}(a) \to P(a)$, and then apply well-foundedness to conclude $\prod_{(a:A)} P(a)$. In other words, if from $\forall(b : A).\, (b < a) \to P(b)$ we can prove $P(a)$, then $\forall(a : A).\, P(a)$. This is called **well-founded induction**.

Lemma 10.1.4. *Well-foundedness is a mere property.*

Proof. Well-foundedness of $<$ is the type $\prod_{(a:A)} \mathsf{acc}(a)$, which is a mere proposition since each $\mathsf{acc}(a)$ is. □

Example 10.1.5. Perhaps the most familiar well-founded relation is the usual strict ordering on $\mathbb{N}$. To show that this is well-founded, we must show that n is accessible for each $n : \mathbb{N}$. This is just the usual proof of "strong induction" from ordinary induction on $\mathbb{N}$.

Specifically, we prove by induction on $n : \mathbb{N}$ that k is accessible for all $k \leq n$. The base case is just that 0 is accessible, which is vacuously true since nothing is strictly less than 0. For the inductive step, we assume that k is accessible for all $k \leq n$, which is to say for all $k < n + 1$; hence by definition $n + 1$ is also accessible.

A different relation on $\mathbb{N}$ which is also well-founded is obtained by setting only $n < \mathsf{succ}(n)$ for all $n : \mathbb{N}$. Well-foundedness of this relation is almost exactly the ordinary induction principle of $\mathbb{N}$.

Example 10.1.6. Let $A : \mathsf{Set}$ and $B : A \to \mathsf{Set}$ be a family of sets. Recall from original reference 5.3 that the W-type $\mathsf{W}_{(a:A)} B(a)$ is inductively generated by the single constructor

- $\mathsf{sup} : \prod_{(a:A)}(B(a) \to \mathsf{W}_{(x:A)} B(x)) \to \mathsf{W}_{(x:A)} B(x)$

We define the relation $<$ on $\mathsf{W}_{(x:A)} B(x)$ by recursion on its second argument:

- For any $a : A$ and $f : B(a) \to \mathsf{W}_{(x:A)} B(x)$, we define $w < \mathsf{sup}(a, f)$ to mean that there merely exists a $b : B(a)$ such that $w = f(b)$.

Now we prove that every $w : \mathsf{W}_{(x:A)} B(x)$ is accessible for this relation, using the usual induction principle for $\mathsf{W}_{(x:A)} B(x)$. This means we assume given $a : A$ and $f : B(a) \to \mathsf{W}_{(x:A)} B(x)$, and also a lifting $f' : \prod_{(b:B(a))} \mathsf{acc}(f(b))$. But then by definition of $<$, we have $\mathsf{acc}(w)$ for all $w < \mathsf{sup}(a, f)$; hence $\mathsf{sup}(a, f)$ is accessible.

Well-foundedness allows us to define functions by recursion and prove statements by induction, such as for instance the following. Recall from original reference 3.5 that $\mathcal{P}(B)$ denotes the *power set* $\mathcal{P}(B) :\equiv (B \to \mathsf{Prop})$.

Lemma 10.1.7. *Suppose B is a set and we have a function*

$$g : \mathcal{P}(B) \to B$$

Then if $<$ is a well-founded relation on A, there is a function $f : A \to B$ such that for all $a : A$ we have

$$f(a) = g\Big(\{\, f(a') \mid a' < a \,\} \Big).$$

(We are using the notation for images of subsets from original reference 10.1.2.)

Proof. We first define, for every $a : A$ and $s : \mathsf{acc}(a)$, an element $\bar{f}(a, s) : B$. By induction, it suffices to assume that s is a function assigning to each $a' < a$ a witness $s(a') : \mathsf{acc}(a')$, and that moreover for each such a' we have an element $\bar{f}(a', s(a')) : B$. In this case, we define

$$\bar{f}(a, s) :\equiv g\Big(\{\, \bar{f}(a', s(a')) \;\big|\; a' < a \,\} \Big).$$

Now since $<$ is well-founded, we have a function $w : \prod_{(a:A)} \mathsf{acc}(a)$. Thus, we can define $f(a) :\equiv \bar{f}(a, w(a))$. □

In classical logic, well-foundedness has a more well-known reformulation. In the following, we say that a subset $B : \mathcal{P}(A)$ is **nonempty** if it is unequal to the empty subset $(\lambda x.\, \bot) : \mathcal{P}(X)$. We leave it to the reader to verify that assuming excluded middle, this is equivalent to mere inhabitation, i.e. to the condition $\exists(x : A).\, x \in B$.

Lemma 10.1.8. *Assuming excluded middle, $<$ is well-founded if and only if every nonempty subset $B : \mathcal{P}(A)$ merely has a minimal element.*

Proof. Suppose first $<$ is well-founded, and suppose $B \subseteq A$ is a subset with no minimal element. That is, for any $a : A$ with $a \in B$, there merely exists a $b : A$ with $b < a$ and $b \in B$.

We claim that for any $a : A$ and $s : \mathsf{acc}(a)$, we have $a \notin B$. By induction, we may assume s is a function assigning to each $a' < a$ a proof $s(a') : \mathsf{acc}(a')$, and that moreover for each such a' we have $a' \notin B$. If $a \in B$, then by assumption, there would merely exist a $b < a$ with $b \in B$, which contradicts this assumption. Thus, $a \notin B$; this completes the induction. Since $<$ is well-founded, we have $a \notin B$ for all $a : A$, i.e. B is empty.

Now suppose each nonempty subset merely has a minimal element. Let $B = \{\, a : A \mid \neg\mathsf{acc}(a) \,\}$. Then if B is nonempty, it merely has a minimal element. Thus there merely exists an $a : A$ with $a \in B$ such that for all $b < a$, we have $\mathsf{acc}(b)$. But then by definition (and induction on truncation), a is merely accessible, and hence accessible, contradicting $a \in B$. Thus, B is empty, so $<$ is well-founded. □

Definition 10.1.9. A well-founded relation $<$ on a set A is **extensional** if for any $a, b : A$, we have

$$\Big(\forall(c : A).\, (c < a) \Leftrightarrow (c < b)\Big) \to (a = b).$$

Note that since A is a set, extensionality is a mere proposition. This notion of "extensionality" is unrelated to function extensionality, and also unrelated to the extensionality of identity types. Rather, it is a "local" counterpart of the axiom of extensionality in classical set theory.

Theorem 10.1.10. *The type of extensional well-founded relations is a set.*

Proof. By the univalence axiom, it suffices to show that if $(A, <)$ is extensional and well-founded and $f : (A, <) \cong (A, <)$, then $f = \mathsf{id}_A$. We prove by induction on $<$ that $f(a) = a$ for all $a : A$. The inductive hypothesis is that for all $a' < a$, we have $f(a') = a'$.

Now since A is extensional, to conclude $f(a) = a$ it is sufficient to show

$$\forall(c : A).\, (c < f(a)) \Leftrightarrow (c < a).$$

However, since f is an automorphism, we have $(c < a) \Leftrightarrow (f(c) < f(a))$. But $c < a$ implies $f(c) = c$ by the inductive hypothesis, so $(c < a) \to (c < f(a))$. On the other hand, if $c < f(a)$, then $f^{-1}(c) < a$, and so $c = f(f^{-1}(c)) = f^{-1}(c)$ by the inductive hypothesis again; thus $c < a$. Therefore, we have $(c < a) \Leftrightarrow (c < f(a))$ for any $c : A$, so $f(a) = a$. □

Definition 10.1.11. If $(A, <)$ and $(B, <)$ are extensional and well-founded, a **simulation** is a function $f : A \to B$ such that

(i) if $a < a'$, then $f(a) < f(a')$, and
(ii) for all $a : A$ and $b : B$, if $b < f(a)$, then there merely exists an $a' < a$ with $f(a') = b$.

Lemma 10.1.12. *Any simulation is injective.*

Proof. We prove by double well-founded induction that for any $a, b : A$, if $f(a) = f(b)$ then $a = b$. The inductive hypothesis for $a : A$ says that for any $a' < a$, and any $b : B$, if $f(a') = f(b)$ then $a = b$. The inner inductive hypothesis for $b : A$ says that for any $b' < b$, if $f(a) = f(b')$ then $a = b'$.

Suppose $f(a) = f(b)$; we must show $a = b$. By extensionality, it suffices to show that for any $c : A$ we have $(c < a) \Leftrightarrow (c < b)$. If $c < a$, then $f(c) < f(a)$ by Definition 10.1.11(i). Hence $f(c) < f(b)$, so by Definition 10.1.11(ii) there merely exists $c' : A$ with $c' < b$ and $f(c) = f(c')$. By the inductive hypothesis for a, we have $c = c'$, hence $c < b$. The dual argument is symmetrical. □

In particular, this implies that in Definition 10.1.11(ii) the word "merely" could be omitted without change of sense.

Corollary 10.1.13. *If $f : A \to B$ is a simulation, then for all $a : A$ and $b : B$, if $b < f(a)$, there* purely *exists an $a' < a$ with $f(a') = b$.*

Proof. Since f is injective, $\sum_{(a:A)}(f(a) = b)$ is a mere proposition. □

We say that a subset $C : \mathcal{P}(B)$ is an **initial segment** if $c \in C$ and $b < c$ imply $b \in C$. The image of a simulation must be an initial segment, while the inclusion of any initial segment is a simulation. Thus, by univalence, every simulation $A \to B$ is *equal* to the inclusion of some initial segment of B.

Theorem 10.1.14. *For a set A, let $P(A)$ be the type of extensional well-founded relations on A. If $<_A : P(A)$ and $<_B : P(B)$ and $f : A \to B$, let $H_{<_A <_B}(f)$ be the mere proposition that f is a simulation. Then (P, H) is a standard notion of structure over $\mathcal{S}et$ in the sense of original reference 9.8.*

Proof. We leave it to the reader to verify that identities are simulations, and that composites of simulations are simulations. Thus, we have a notion of structure. For standardness, we must show that if $<$ and $\prec$ are two extensional well-founded relations on A, and id_A is a simulation in both directions, then $<$ and $\prec$ are equal. Since extensionality and well-foundedness are mere propositions, for this it suffices to have $\forall(a, b : A).\, (a < b) \Leftrightarrow (a \prec b)$. But this follows from Definition 10.1.11(i) for id_A. □

Corollary 10.1.15. *There is a category whose objects are sets equipped with extensional well-founded relations, and whose morphisms are simulations.*

In fact, this category is a poset.

Lemma 10.1.16. *For extensional and well-founded $(A, <)$ and $(B, <)$, there is at most one simulation $f : A \to B$.*

Proof. Suppose $f, g : A \to B$ are simulations. Since being a simulation is a mere property, it suffices to show $\forall(a : A).\, (f(a) = g(a))$. By induction on $<$, we may suppose $f(a') = g(a')$ for all $a' < a$. And by extensionality of B, to have $f(a) = g(a)$ it suffices to have $\forall(b : B).\, (b < f(a)) \Leftrightarrow (b < g(a))$.

But since f is a simulation, if $b < f(a)$, then we have $a' < a$ with $f(a') = b$. By the inductive hypothesis, we have also $g(a') = b$, hence $b < g(a)$. The dual argument is symmetrical. □

Thus, if A and B are equipped with extensional and well-founded relations, we may write $A \leq B$ to mean there exists a simulation $f : A \to B$. Corollary 10.1.15 implies that if $A \leq B$ and $B \leq A$, then $A = B$.

Definition 10.1.17. An **ordinal** is a set A with an extensional well-founded relation which is *transitive*, i.e. satisfies $\forall(a, b, c : A).\, (a < b) \to (b < c) \to (a < c)$.

Example 10.1.18. Of course, the usual strict order on $\mathbb{N}$ is transitive. It is easily seen to be extensional as well; thus it is an ordinal. As usual, we denote this ordinal by ω.

Let Ord denote the type of ordinals. By the previous results, Ord is a set and has a natural partial order. We now show that Ord also admits a well-founded relation.

If A is an ordinal and $a : A$, let $A_{/a} :\equiv \{\, b : A \mid b < a \,\}$ denote the initial segment. Note that if $A_{/a} = A_{/b}$ as ordinals, then that isomorphism must respect their inclusions into A (since simulations form a poset), and hence they are equal as subsets of A. Therefore, since A is extensional, $a = b$. Thus the function $a \mapsto A_{/a}$ is an injection $A \to \mathsf{Ord}$.

Definition 10.1.19. For ordinals A and B, a simulation $f : A \to B$ is said to be **bounded** if there exists $b : B$ such that $A = B_{/b}$.

The remarks above imply that such a b is unique when it exists, so that boundedness is a mere property.

We write $A < B$ if there exists a bounded simulation from A to B. Since simulations are unique, $A < B$ is also a mere proposition.

Theorem 10.1.20. $(\mathsf{Ord}, <)$ *is an ordinal.*

Proof. Let A be an ordinal; we first show that $A_{/a}$ is accessible (in Ord) for all $a : A$. By well-founded induction on A, suppose $A_{/b}$ is accessible for all $b < a$. By definition of accessibility, we must show that B is accessible in Ord for all $B < A_{/a}$. However, if $B < A_{/a}$ then there is some $b < a$ such that $B = (A_{/a})_{/b} = A_{/b}$, which is accessible by the inductive hypothesis. Thus, $A_{/a}$ is accessible for all $a : A$.

Now to show that A is accessible in Ord, by definition we must show B is accessible for all $B < A$. But as before, $B < A$ means $B = A_{/a}$ for some $a : A$, which is accessible as we just proved. Thus, Ord is well-founded.

For extensionality, suppose A and B are ordinals such that $\prod_{(C:\mathsf{Ord})}(C < A) \Leftrightarrow (C < B)$. Then for every $a : A$, since $A_{/a} < A$, we have $A_{/a} < B$, hence there is $b : B$ with $A_{/a} = B_{/b}$. Define $f : A \to B$ to take each a to the corresponding b; it is straightforward to verify that f is an isomorphism. Thus $A \cong B$, hence $A = B$ by univalence.

Finally, it is easy to see that $<$ is transitive. □